

Notes: Bertrand, Goldin & Katz (AEJ Applied Economics, 2010)

Dynamics of the Gender Gap for Young Professionals in the Financial and Corporate Sectors

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1 Big picture

- **Question.** Why do earnings gaps between male and female elite MBAs open up so quickly after graduation, even when men and women start from very similar jobs and very similar initial earnings?
- **Setting.** The paper studies full-time MBA graduates from the University of Chicago Booth School of Business, graduation cohorts 1990–2006, using a retrospective survey matched to administrative school records.
- **Main fact.** Male and female MBAs have nearly equal earnings at career start, but the male earnings advantage reaches about 30 log points after five years and almost 60 log points 10–16 years after MBA completion.
- **Three proximate drivers.** The paper attributes most of the growing gap to: MBA training differences, career interruptions, and weekly hours worked.
- **Human capital at entry.** Men take more finance courses and have somewhat higher MBA GPAs. These differences are small, but finance coursework and MBA performance have high labor-market returns.
- **Career interruptions.** Women, especially mothers, are much more likely to have nonwork spells of at least six months. Career interruptions are heavily penalized in high-powered finance and corporate jobs.
- **Hours.** Women work fewer weekly hours over time. The decline is much stronger for women with children. The earnings penalty to shorter hours is large because the relevant jobs reward long and continuous work.
- **Motherhood.** Children are the main source of women’s greater career discontinuity and lower hours. Women without children look much closer to men in labor supply and experience.
- **Spousal income.** Mothers married to high-earning spouses reduce labor supply much more after a first birth than mothers with lower-earning spouses.
- **Interpretive stance.** The evidence is best read as a proximate accounting and dynamic event-study analysis, not as a randomized causal decomposition of discrimination versus preferences versus constraints.
- **Economic takeaway.** In elite corporate and financial labor markets, small differences in training and especially small deviations from continuous, long-hour employment can cumulate into very large gender earnings gaps.

2 Data and measurement

2.1 MBA survey and administrative records

- The authors conducted a web survey of University of Chicago full-time MBA graduates from the classes of 1990–2006.
- The survey was fielded from November 2006 to June 2007.
- Part-time and executive MBA graduates were excluded.
- The survey asked respondents about every post-MBA job stage lasting at least six months: earnings, weekly hours, job function, sector, firm type, firm size, reasons for choosing the job, and reasons for leaving.
- It also asked about every post-MBA nonemployment spell of at least six months and the reasons for the spell.
- The survey data were matched to school administrative records on courses, grades, undergraduate institution, undergraduate GPA, GMAT scores, age, ethnicity, and citizenship.
- Of the matched respondents, the core sample contains 2,485 MBAs: 1,856 men and 629 women.

Interpretation: The paper’s advantage is that it observes both pre-market human capital and detailed post-MBA career histories. This lets the authors separate entry differences from later dynamic labor-supply differences.

2.2 Construction of the individual-year panel

- Survey responses are converted into an unbalanced person-year panel.
- Earnings in a year are interpolated from the first and last year of each job stage.
- Earnings are total annual earnings before taxes and deductions, including salary and bonus.
- Earnings bins are converted into midpoints and deflated to 2006 dollars.
- Weekly hours are also reported in bins and converted into numerical hours.
- A “career interruption” is a spell of six months or more without paid work.

Interpretation: The annual panel structure makes it possible to track how gender gaps evolve with years since MBA and around major family events such as the first birth.

2.3 Key labor-market outcomes

- **Annual earnings:** total salary and bonus in 2006 dollars.
- **Log annual earnings:** the main wage outcome for employed respondents.
- **Not working:** an indicator for not working in a given year.
- **Actual post-MBA experience:** years of work experience accumulated after MBA completion.
- **Cumulative years not working:** total time spent in nonwork spells after MBA completion.
- **Weekly hours:** usual hours worked in the current job.
- **Full-time/full-year work:** 52 weeks of work and more than 30–40 hours per week.

Interpretation: The paper treats earnings as the ultimate outcome but emphasizes that the earnings gap is built through labor-supply and experience dynamics.

2.4 Family variables

- The survey records marital status and spouse education, employment, and earnings.
- It records whether respondents have children, the birth year of each child, and childcare allocation.
- The main family indicator is whether a woman has at least one child in a given year.
- Spousal earnings are grouped into low, medium, and high categories, with high earnings generally above \$200,000 in the survey year.

Interpretation: The family variables are central because the largest labor-supply differences are concentrated among MBA mothers, especially those with high-earning spouses.

2.5 Pre-MBA and MBA controls

- Pre-MBA controls include citizenship, race/ethnicity, undergraduate institution ranking, undergraduate GPA, GMAT scores, pre-MBA industry, and pre-MBA job function.
- MBA performance includes overall MBA GPA and the fraction of MBA classes taken in finance.
- Job-setting controls include job function and employer type.
- Job-choice controls include reported reasons such as career advancement, compensation, culture, flexible hours, remote work, limited travel, and location.

Interpretation: These controls make the analysis a sequential decomposition: how much of the raw gender gap remains after adding human capital, labor supply, preferences over job attributes, and job setting?

3 Models and empirical specifications

3.1 Descriptive earnings-gap profile

The paper first estimates gender gaps by years since MBA completion using regressions of the form

$$Y_{it} = \alpha + \sum_k \beta_k (Female_i \times 1\{YearsSinceMBA_{it} = k\}) + \Gamma_{cohort(i),t} + \varepsilon_{it}, \quad (1)$$

where Y_{it} is an earnings or labor-supply outcome, and $\Gamma_{cohort(i),t}$ are cohort-by-year dummies.

Interpretation: The coefficients β_k trace how the male-female gap changes as the MBA cohort ages, net of common cohort-year effects.

3.2 Pooled wage regressions

The main pooled wage specification is

$$\log(Earnings_{it}) = \alpha + \beta Female_i + X'_{it}\theta + \Gamma_{cohort(i),t} + u_{it}. \quad (2)$$

The vector X_{it} is expanded column by column to include:

- pre-MBA characteristics;
- MBA GPA and finance-course share;
- actual post-MBA experience and its square;
- an indicator for any nonwork spell;
- weekly-hours dummies;
- reasons for choosing the current job;
- job function and employer type.

Interpretation: The coefficient β is not interpreted as a pure discrimination parameter. It is the remaining gender gap after a particular set of observables has been included.

3.3 Gender gap by years since MBA

To understand timing, the paper repeats the wage decomposition separately by years since MBA:

$$\log(Earnings_{it}) = \alpha + \beta_k Female_i + X'_{it}\theta + \Gamma_{cohort(i),t} + u_{it} \quad \text{within year-since-MBA group } k. \quad (3)$$

Interpretation: This shows which controls explain the gap at entry, at mid-career, and after a decade or more. Labor-market experience and hours become increasingly important with time since graduation.

3.4 Labor-supply regressions by children and spouse income

The paper estimates outcomes such as not working, post-MBA experience, and log weekly hours using specifications like

$$Y_{it} = \alpha + \beta_1 FemaleWithChild_{it} + \beta_2 FemaleWithoutChild_{it} + X'_{it}\theta + \Gamma_{cohort(i),t} + u_{it}. \quad (4)$$

It then interacts motherhood with spouse-income groups:

$$Y_{it} = \alpha + \beta_1 Mother_{it} + \beta_2 (Mother_{it} \times HighSpouse_i) + \beta_3 (Mother_{it} \times MediumSpouse_i) + \dots + u_{it}. \quad (5)$$

Interpretation: These regressions ask whether the gender gap is really a motherhood gap, and whether it is amplified by household income structure.

3.5 First-birth event-study specification

For the dynamic analysis around first birth, the paper estimates person-fixed-effects models:

$$Y_{it} = \alpha_i + \Gamma_{cohort(i),t} + f(Age_{it}) + \sum_{r \in \mathcal{R}} \delta_r 1\{t - BirthYear_i = r\} + \varepsilon_{it}, \quad (6)$$

where r denotes event-time bins: one or two years before birth, birth year, one or two years after birth, three or four years after, and five or more years after.

Interpretation: The fixed effects compare the same individual's labor-market outcomes before and after the first birth. The benchmark period is three or more years before the first birth.

3.6 Decomposition logic

The paper uses estimated coefficients and gender differences in covariates to decompose the earnings gap:

$$\widehat{Gap} = \widehat{\beta}_{Female} + (\bar{X}_m - \bar{X}_f)' \widehat{\theta}. \quad (7)$$

Interpretation: This is an accounting exercise. It asks how much of the observed male-female earnings difference is associated with measured differences in skills, work history, hours, job attributes, and job settings.

4 Identification and empirical design

4.1 What the paper identifies

- The paper identifies proximate determinants of the gender earnings gap among elite MBAs.

- It does not randomly assign children, hours, job preferences, or career interruptions.
- The estimates show that measured human capital and labor-supply variables account for most of the observed gap.
- The event-study models around first birth identify within-person changes in outcomes relative to pre-birth years.

Interpretation: The paper is strongest as a dynamic descriptive and accounting analysis. It tells us where the gap comes from in the data, not the ultimate social or biological cause of every choice.

4.2 Why the setting is useful

- The sample is highly selected and homogeneous: all respondents are elite MBAs from one top school.
- Men and women start with similar degrees, similar initial labor-force attachment, and nearly equal initial earnings.
- Administrative records reduce measurement error in pre-market human capital.
- The survey records job histories, hours, nonwork spells, family timing, and reasons for job choices.

Interpretation: Because the sample is so elite and initially similar, large later gaps are difficult to attribute simply to broad differences in education or initial labor-market access.

4.3 Main threats and limitations

- The survey response rate is about 31 percent, although respondents and nonrespondents appear similar on many observables.
- Earnings and hours are retrospective and binned, so they require interpolation and midpoint conversion.
- Children and spousal income are endogenous household choices.
- Job preferences reported in the survey may partly reflect constrained choices or workplace norms.
- The sample is one elite MBA program, so external validity to all workers is limited.
- The paper cannot fully distinguish preferences, discrimination, social norms, household bargaining, and firm policies.

Interpretation: The evidence is powerful for documenting the anatomy of the gap, but it does not prove that the observed choices are unconstrained or fully voluntary.

5 Tables and figures

This section follows the same compact “Read” plus “Economic message” format. Adjacent visuals are grouped where possible, and the images are directly cropped from the paper.

5.1 Table 1 and Figure 1: labor supply and earnings paths

Read:

- Table 1 shows that early labor-force participation is high for both genders, but the gender gap widens sharply over time.

- At year 9, 12.9 percent of women are not working compared with 1.1 percent of men; at 10 or more years, 16.6 percent of women are not working compared with 1.0 percent of men.
- Women’s cumulative years not working reach 1.052 by 10 or more years after MBA completion, compared with 0.120 for men.
- Women’s mean weekly hours fall from 59.1 at graduation to 49.3 at 10 or more years, while men’s fall from 60.9 to 56.7.
- Figure 1 shows earnings diverging rapidly. Female medians flatten early, while male means and especially male 90th percentile salaries keep rising.

Economic message: Initial equality is not persistent. The gender gap is a dynamic career-path phenomenon built through employment, hours, and earnings growth.

TABLE 1—LABOR SUPPLY BY GENDER AND NUMBER OF YEARS SINCE MBA GRADUATION: DESCRIPTIVE STATISTICS

	Number of years since MBA graduation					
	0	1	3	6	9	≥ 10
Share not working at all in current year						
Female	0.054	0.012	0.027	0.067	0.129	0.166
Male	0.028	0.005	0.003	0.008	0.011	0.010
Share working full time/full-year (52 weeks and > 30 to 40 hours per week)						
Female	NA	0.89	0.84	0.78	0.69	0.62
Male	NA	0.93	0.94	0.93	0.93	0.92
Cumulative share with any no work spell (until given year)						
Female	0.064	0.088	0.143	0.229	0.319	0.405
Male	0.032	0.040	0.064	0.081	0.095	0.101
Cumulative years not working						
Female	0	0.050	0.118	0.282	0.569	1.052
Male	0	0.026	0.045	0.069	0.098	0.120
Mean weekly hours worked for the employed						
Female	59.1	58.8	56.2	54.7	51.5	49.3
Male	60.9	60.7	59.5	57.9	57.5	56.7
Share working part time (≤ 30 to 40 hours per week)						
Female	0.04	0.05	0.07	0.09	0.15	0.22
Male	0.02	0.02	0.02	0.03	0.03	0.04
Share working fewer than 52 weeks						
Female	NA	0.07	0.07	0.09	0.06	0.06
Male	NA	0.05	0.04	0.03	0.03	0.03

Note: Individuals who do not work at all in a given year are excluded from those “working part time” and “working fewer than 52 weeks” and are included as zeros in the definition of “working full time/full year.”

(a) Table 1: labor supply by gender and years since MBA

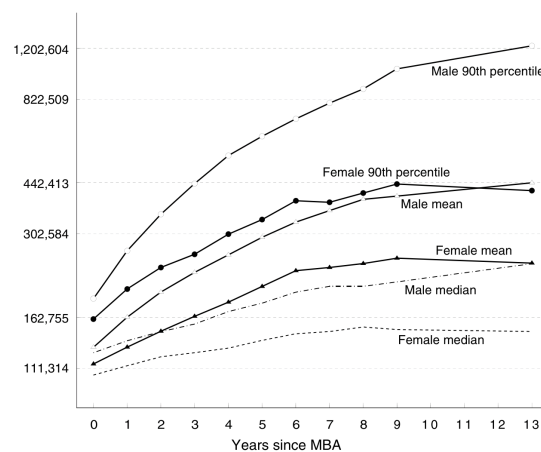


FIGURE 1. MALE AND FEMALE MEAN, MEDIAN, AND NINETIETH PERCENTILE ANNUAL SALARIES (2006 DOLLARS) BY YEARS SINCE MBA

(b) Figure 1: male and female salary paths

5.2 Tables 2 and 3: raw gaps and pooled wage regressions

Read:

- Table 2 shows the earnings gap grows with time since MBA: log annual earnings gap is 0.113 at year 0, 0.260 at year 3, 0.402 at year 9, and 0.583 at 10 or more years.
- The labor-supply gaps also grow: cumulative years not working, probability of not working, annual weeks worked, annual hours, and log weekly hours all move against women over time.
- Table 3 shows the pooled female coefficient is -0.287 with only cohort-by-year controls.
- Adding pre-MBA characteristics and MBA performance reduces it to -0.190 .
- Adding hours, experience, and nonwork spells reduces the coefficient further; in the preferred column with MBA and labor-supply controls, the female coefficient is about -0.064 .
- Adding reasons for job choice, job function, and employer type makes the residual female coefficient small and statistically insignificant.

Economic message: Most of the raw gender earnings gap is associated with measured MBA performance, finance coursework, work experience, interruptions, and hours rather than a large unexplained residual.

TABLE 2—GENDER GAP IN EARNINGS AND LABOR SUPPLY

Number of years since receipt of MBA	Annual earnings (1)	Log (annual earnings) (2)	Log (annual earnings) first in job (3)	Cumulative years not working (4)	Not working at all in current year (5)	Annual hours worked in current year* (6)	Annual hours worked in current year* (7)	Log (weekly hours worked) (8)
0	-16.943 [2.223]***	-0.113 [0.018]***	-0.110 [0.018]***	-0.001 [0.000]**	0.024 [0.009]**	-1.628 [0.530]***	-181.9 [46.7]***	-0.034 [0.012]***
1	-31.083 [3.999]***	-0.170 [0.022]***	-0.174 [0.059]***	0.022 [0.008]***	-0.907 [0.004]	-111.7 [0.357]**	-0.036 [0.012]***	
3	-65.799 [9.238]***	-0.260 [0.033]***	-0.189 [0.075]**	0.070 [0.017]**	0.023 [0.007]***	-2.059 [0.512]***	-260.9 [46.0]***	-0.074 [0.015]**
6	-97.662 [21.093]***	-0.320 [0.048]***	-0.272 [0.125]**	0.212 [0.041]***	0.059 [0.013]***	-4.665 [0.812]***	-363.5 [61.2]***	-0.083 [0.019]***
9	-152.002 [31.672]***	-0.402 [0.066]***	-0.360 [0.170]**	0.476 [0.079]***	0.116 [0.021]***	-7.029 [1.136]***	-622.6 [77.3]***	-0.138 [0.026]***
≥ 10	-195.576 [40.295]***	-0.583 [0.084]***	-0.604 [0.112]***	0.925 [0.143]***	0.155 [0.025]***	-8.812 [1.345]***	-789.9 [96.2]***	-0.195 [0.035]***
Cohort × year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,161	20,161	5,220	21,290	21,290	21,286	20,925	20,430
R ²	0.11	0.15	0.21	0.14	0.08	0.08	0.09	0.07

Notes: The unit of observation is a survey respondent in a given post-MBA year. Each column corresponds to a different regression and each includes (cohort × year) dummies and interactions between a female dummy and dummy variables for the number of years since receipt of the MBA. The table reports the estimated coefficients on the interaction terms between the female dummy and years since receiving the MBA. "Annual earnings" is defined as total earnings before taxes and other deductions, including salary and bonus, and is coded as missing when the individual is not working. "Hourly wage" is computed by dividing annual earnings by (weekly hours worked × 52). Columns 1–3 only include those with positive earnings. Column 3 includes only the first year at a given job. Standard errors (in brackets) are clustered at the individual level.

* Including zeros

*** Significant at the 1 percent level.

** Significant at the 5 percent level

TABLE 3—WAGE REGRESSIONS FOR POOLED SAMPLE

	Dependent variable: Log (annual earnings)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	-0.287 [0.035]***	-0.190 [0.033]***	-0.146 [0.032]***	-0.173 [0.030]***	-0.094 [0.029]***	-0.064 [0.029]**	-0.054 [0.028]	-0.038 [0.025]
MBA GPA		0.429 [0.054]***	0.406 [0.053]***		0.369 [0.051]***	0.351 [0.051]***	0.367 [0.049]***	0.347 [0.043]***
Fraction finance classes		1.833 [0.211]***	1.807 [0.206]***		1.758 [0.199]***	1.737 [0.194]***	1.65 [0.193]***	0.430 [0.180]**
Actual post-MBA exp			0.046 [0.075]			0.085 [0.071]	0.056 [0.068]	0.029 [0.064]
Actual post-MBA exp ²			0.010 [0.004]***			0.005 [0.004]	0.008 [0.003]**	0.007 [0.003]**
Any no work spell			-0.290 [0.067]***			-0.228 [0.062]***	-0.218 [0.061]***	-0.173 [0.054]***
Dummy variables:								
Weekly hours worked	No	No	No	Yes	Yes	Yes	Yes	Yes
Pre-MBA characteristics	No	Yes	Yes	No	Yes	Yes	Yes	Yes
Reason for choosing job	No	No	No	No	No	No	Yes	Yes
Job function	No	No	No	No	No	No	No	Yes
Employer type	No	No	No	No	No	No	No	Yes
Cohort × year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	12.156 [0.018]***	9.493 [0.585]***	8.809 [0.667]***	10.385 [0.151]***	8.08 [0.603]***	7.525 [0.694]***	8.229 [0.733]***	8.324 [0.547]***
Observations	18,272	18,272	18,272	18,272	18,272	18,272	18,272	18,272
R ²	0.15	0.31	0.34	0.26	0.40	0.41	0.43	0.54

5.3 Tables 4 and 5: timing of the gap and the role of children

Read:

- Table 4 shows the raw gender wage gap rises from 8.9 log points at graduation to 56.5 log points at 10 or more years.
- Pre-MBA characteristics and MBA performance explain part of the gap, especially early on.
- The biggest compression comes from adding labor-market experience and weekly hours worked.
- After adding weekly hours, the 10-plus year gap falls from 31.2 log points to 9.8 log points.
- Table 5 shows that women with children have much lower labor supply than men: they are 20 percentage points more likely not to work, have 0.66 fewer years of post-MBA experience, and work 23.8 log points fewer weekly hours.
- Women without children are much closer to men, especially in hours worked.

Economic message: The gap is not just a gender gap; much of it is a motherhood and career-continuity gap that becomes visible several years after MBA completion.

TABLE 4—GENDER WAGE GAP BY YEARS SINCE MBA

	Number of years since MBA receipt					
	0	1	3	6	9	≥ 10
1. With no controls	-0.089 [0.020]***	-0.154 [0.025]***	-0.253 [0.038]***	-0.308 [0.056]***	-0.376 [0.079]***	-0.565 [0.045]***
With controls:						
2. Pre-MBA characteristics	-0.080 [0.021]***	-0.136 [0.026]***	-0.204 [0.039]***	-0.248 [0.057]***	-0.320 [0.084]***	-0.479 [0.045]***
3. Add MBA performance	-0.054 [0.021]***	-0.103 [0.025]***	-0.154 [0.037]***	-0.180 [0.055]***	-0.257 [0.082]***	-0.446 [0.044]***
4. Add labor market exp.	-0.053 [0.021]**	-0.093 [0.025]***	-0.134 [0.037]***	-0.143 [0.055]***	-0.181 [0.082]**	-0.312 [0.044]***
5. Add weekly hours worked	-0.036 [0.020]	-0.073 [0.023]***	-0.073 [0.036]**	-0.079 [0.053]	-0.047 [0.078]	-0.098 [0.042]**
6. Add reason for choosing job	-0.033 [0.020]	-0.067 [0.023]***	-0.064 [0.035]	-0.075 [0.053]	-0.031 [0.079]	-0.066 [0.042]
7. Add job setting characteristics	-0.025 [0.019]	-0.060 [0.022]***	-0.064 [0.032]**	-0.080 [0.048]	0.002 [0.071]	-0.010 [0.037]

Notes: The dependent variable in each equation is log (annual earnings). Each cell corresponds to a different regression. The unit of observation is a survey respondent in a given post-MBA year. In the regression without any controls, the sample is restricted to those survey respondents with non-missing pre-MBA characteristics. All regressions include (cohort × year) dummies and a female dummy. In each column, the sample is restricted to (individual × year) observations that correspond to the number of years since graduation listed in that column. Each cell gives the estimated coefficient on the female dummy. The controls for Pre-MBA characteristics, Weekly Hours Worked, and Reason for Choosing a Job are the same as those described in the notes to Table 3. MBA performance includes overall MBA GPA and fraction of finance classes. Labor market experience includes a quadratic in actual experience since MBA graduation, a dummy variable for "any no work spell" and dummy variables for weekly hours worked. Job setting characteristics include job function dummies and "Employer type" dummies. "Employer type" dummies include: Public for-profit, < 100 employees; Public for-profit, 100–1,000 employees; Public for-profit, 1,000–15,000 employees; Private for-profit, > 15,000 employees; Private for-profit, < 100 employees; Private for-profit, 101–1,000 employees; Private for-profit, 1,000–15,000 employees; Private for-profit, > 15,000 employees.

(a) Table 4: wage gap by years since MBA

TABLE 5—DETERMINANTS OF THE GENDER GAP IN LABOR SUPPLY: THE ROLE OF CHILDREN

	Actual post-MBA experience				Log (weekly hours worked)	
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
Female	0.084 [0.009]***		-0.286 [0.039]***		-0.089 [0.013]***	
Female with child		0.200 [0.024]***		-0.660 [0.094]***		-0.238 [0.031]***
Female without child		0.034 [0.007]***		-0.126 [0.031]***		-0.033 [0.012]***
Pre-MBA characteristics	Yes	Yes	Yes	Yes	Yes	Yes
MBA performance	Yes	Yes	Yes	Yes	Yes	Yes
Cohort × year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-0.175 [0.145]	-0.111 [0.126]	5.929 [0.618]***	5.757 [0.550]***	3.951 [0.462]***	3.914 [0.426]***
Observations	19,366	19,286	19,366	19,286	18,611	18,535
R ²	0.07	0.11	0.98	0.98	0.14	0.16

Notes: The unit of observation is a survey respondent in a given post-MBA year. "Female with child" ("Female without child") is a dummy variable that equals 1 if the respondent is a female and has at least one child (no child) in that year. Pre-MBA characteristics include: a dummy for US citizen, a white dummy, an Asian dummy, a dummy for "top 10" undergraduate institution, a dummy for "top 10–20" undergraduate institution, undergraduate GPA, a dummy for missing undergraduate GPA, a quadratic in age, verbal GMAT score, quantitative GMAT score, a dummy for pre-MBA industry and a dummy for pre-MBA job function. MBA performance includes overall MBA GPA and fraction of finance classes. Standard errors (in brackets) are clustered at the individual level.

*** Significant at the 1 percent level.

(b) Table 5: children and labor supply

5.4 Tables 6 and 7: spousal income and selection

Read:

- Table 6 shows motherhood has much larger labor-supply effects for women with high-earning spouses.
- A mother with a high-earning spouse is about 30 percentage points less likely to work than the average man: the base motherhood coefficient is 0.119 and the high-spouse interaction is 0.185.
- High-spouse mothers also accumulate more nonemployment and work shorter hours.
- For women without children, higher spousal earnings do not reduce labor supply in the same way.
- Table 7 tests selection into marriage and childbearing using predicted early-career earnings.
- Married women and mothers are not negatively selected in a way that could explain their later earnings losses; if anything, male marriage and childbearing are associated with positive predicted earnings.

Economic message: Household income structure strongly shapes the labor-supply response to children. Selection into motherhood does not explain away the post-birth labor-market slowdown.

TABLE 6—FEMALE LABOR SUPPLY AND SPOUSAL INCOME

	Not working	Actual post-MBA experience	Log (weekly hours worked)
	(1)	(2)	(3)
Female with child	0.119 [0.046]***	−0.420 [0.175]**	−0.169 [0.041]***
Female with child × spouse with high earnings	0.185 [0.061]***	−0.538 [0.240]**	−0.189 [0.078]**
Female with child × spouse with medium earnings	0.026 [0.059]	−0.148 [0.235]	−0.045 [0.063]
Female without child	0.047 [0.021]**	−0.162 [0.087]	−0.067 [0.025]***
Female without child × spouse with high earnings	−0.028 [0.023]	0.122 [0.088]	0.100 [0.034]***
Female without child × spouse with medium earnings	−0.011 [0.023]	0.100 [0.086]	−0.008 [0.030]
Pre-MBA characteristics	Yes	Yes	Yes
MBA performance	Yes	Yes	Yes
Spouse salary dummies	Yes	Yes	Yes
Cohort × year dummies	Yes	Yes	Yes
Constant	−0.138 [0.119]	5.795 [0.520]***	3.891 [0.426]***
Observations	17,655	17,655	17,010
R ²	0.14	0.99	0.18

Notes: The unit of observation is a survey respondent in a given post-MBA year. The sample includes only women who were “married” at the survey date and also excludes women with

(a) Table 6: female labor supply and spousal income

TABLE 7—SELECTIVITY IN MARRIAGE AND CHILDBEARING

	Dependent variable: Predicted log (annual earnings)	
	(1)	(2)
Female, married	−0.066 [0.025]***	
Female, unmarried	−0.084 [0.029]***	
Male, married	0.052 [0.020]***	
Female with child		−0.073 [0.025]***
Female without child		−0.088 [0.022]***
Male with child		0.056 [0.016]***
Constant	−0.061 [0.018]***	−0.053 [0.012]***
Observations	2,317	2,310
R ²	0.02	0.03

Notes: The unit of observation is a survey respondent. Each column corresponds to a different

(b) Table 7: selectivity in marriage and childbearing

5.5 Tables 8 and 9: event studies around first birth

Read:

- Table 8 estimates person-fixed-effects event studies around first birth.
- For women, not working rises sharply in the year of birth and remains elevated thereafter: 9.6 percentage points in the birth year, 13.1 in years 1–2, 17.8 in years 3–4, and 19.0 five or more years after birth.
- Women’s log annual earnings fall by about 16 log points in years 1–2, 29 log points in years 3–4, and 30 log points five or more years after birth.
- Women’s weekly hours decline in the birth year and remain much lower, reaching a 23.3 log point decline five or more years after birth.
- Men do not experience analogous declines around first birth.

- Table 9 shows that the post-birth effect is much stronger for women with spouses earning more than \$200,000.

Economic message: The timing points to childbirth as a major turning point in the careers of female MBAs. The effect is persistent and much stronger in high-income-spouse households.

TABLE 8—IMPACT OF FIRST BIRTH ON EMPLOYMENT STATUS, SALARY, AND WORKING HOURS

	Not working		Log (annual earnings)		Annual earnings (0 if not working)		Log (weekly hours worked)	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Year of birth of first child	-0.001 [0.007]	0.096 [0.032]***	0.008 [0.036]	-0.096 [0.054]	-2.315 [20.942]	-45.666 [20.936]**	-0.006 [0.010]	-0.126 [0.029]***
Years after birth of first child:								
1 or 2	-0.009 [0.007]	0.131 [0.036]***	0.040 [0.040]	-0.164 [0.066]**	5.117 [24.118]	-64.586 [26.335]**	-0.013 [0.011]	-0.168 [0.036]***
3 or 4	-0.007 [0.008]	0.178 [0.045]***	0.065 [0.049]	-0.292 [0.092]***	9.721 [29.915]	-99.397 [34.839]***	-0.011 [0.013]	-0.238 [0.049]***
5 or more	0.000 [0.0012]	0.190 [0.054]***	0.162 [0.060]**	-0.301 [0.119]**	62.581 [37.872]	-101.719 [44.384]**	0.000 [0.017]	-0.233 [0.071]***
Years before birth of first child:								
1 or 2	-0.006 [0.005]	-0.015 [0.021]	-0.008 [0.030]	-0.051 [0.041]	-7.830 [16.303]	-19.137 [15.226]	-0.005 [0.009]	-0.043 [0.023]
Observations	14,490	5,070	13,969	4,545	14,523	5,070	14,193	4,560
R ²	0.29	0.46	0.77	0.73	0.66	0.68	0.72	0.68

Notes: The unit of observation is a survey respondent in a given post-MBA year. Individuals who had children prior to completing their MBA are not included in the regressions. Each column corresponds to a different regression. All regressions include (cohort $\times$ year) dummies, person fixed effects, and a quadratic in age. Each row reports the coefficient on a dummy variable indicating the year of first birth or the number of years after or before the birth of the first child. Standard errors (in brackets) are clustered at the individual level.
***Significant at the 1 percent level.
**Significant at the 5 percent level.

TABLE 9—IMPACT OF BIRTH OF FIRST CHILD ON FEMALE EMPLOYMENT STATUS, SALARY, AND WORKING HOURS: BY SPOUSE'S INCOME LEVEL IN 2006

	Spouse earns at most \$200k in 2006				Spouse earns more than \$200k in 2006			
	Not working (1)	Log (annual earnings) (2)	Log (weekly hours worked) (3)	Annual earnings (0 if not working) (4)	Not working (5)	Log (annual earnings) (6)	Log (weekly hours worked) (7)	Annual earnings (0 if not working) (8)
Year of first birth	0.006 [0.040]	-0.003 [0.089]	-0.141 [0.046]***	-9.659 [23.599]	0.170 [0.070]**	-0.043 [0.099]	-0.088 [0.060]	-89.785 [62.741]
Years after birth of first child:								
1 or 2	0.003 [0.047]	-0.040 [0.102]	-0.162 [0.054]***	-10.100 [28.203]	0.224 [0.081]***	-0.120 [0.124]	-0.158 [0.077]**	-128.393 [78.332]
3 or 4	0.025 [0.064]	-0.100 [0.142]	-0.238 [0.071]***	-13.151 [38.170]	0.281 [0.096]***	-0.264 [0.180]	-0.206 [0.105]	-198.134 [105.466]
5 or more	0.013 [0.087]	0.069 [0.192]	-0.224 [0.099]**	3.236 [49.555]	0.315 [0.107]***	-0.382 [0.233]	-0.194 [0.153]	-216.101 [123.738]
Years before birth of first child:								
1 or 2	-0.066 [0.030]**	-0.047 [0.068]	-0.082 [0.036]**	-1.574 [16.604]	0.018 [0.044]	0.078 [0.074]	0.043 [0.048]	-38.502 [42.786]
Observations	1,841	1,674	1,672	1,841	1,422	1,198	1,190	1,422
R ²	0.50	0.69	0.66	0.64	0.53	0.82	0.77	0.74

Notes: The unit of observation is a female survey respondent in a given post-MBA year. The sample includes those who were married at the survey date and excludes those who had children prior to completing their MBA. Each col-

5.6 Table 10: reasons for not working, choosing jobs, and leaving jobs

Read:

- Table 10 asks whether post-birth changes are related to family reasons or career reasons.
- After a first birth, the probability of not working for family reasons rises strongly, while not working for career reasons does not.
- The probability of choosing a job for family reasons rises after birth.
- The probability of choosing a job for career reasons falls after birth.
- Leaving a job for family reasons is less central than job choice and nonemployment reasons in the table.

Economic message: The evidence points to mothers re-optimizing toward family-compatible jobs and nonemployment rather than simply being pushed out by layoffs or lack of suitable jobs.

TABLE 10—IMPACT OF BIRTH OF FIRST CHILD ON WOMEN’S REASONS FOR NOT WORKING, CHOOSING OR LEAVING A JOB, AND JOB CHARACTERISTICS

	Not working: reasons		Choosing job: reasons		Leave job	Leave job: reasons		
	Family (1)	Career (2)	Family (3)	Career (4)	(5)	Family (6)	Career (7)	Career2 (8)
Year of birth of first child	0.119 [0.030]***	−0.011 [0.008]	0.125 [0.032]***	−0.080 [0.041]	−0.014 [0.035]	0.039 [0.022]	−0.014 [0.014]	−0.019 [0.021]
Years after birth of first child: 1 or 2	0.152 [0.034]***	−0.008 [0.010]	0.197 [0.040]***	−0.132 [0.051]***	−0.007 [0.031]	0.028 [0.017]	−0.019 [0.013]	−0.017 [0.019]
3 or 4	0.200 [0.042]***	−0.009 [0.013]	0.256 [0.050]***	−0.214 [0.061]***	−0.043 [0.034]	0.035 [0.018]	−0.009 [0.016]	−0.017 [0.022]
5 or more	0.233 [0.050]***	−0.018 [0.014]	0.231 [0.062]***	−0.135 [0.074]	−0.065 [0.038]	0.005 [0.019]	0.000 [0.019]	−0.007 [0.025]
Years before birth of first child: 1 or 2	−0.006 [0.016]	−0.004 [0.008]	0.040 [0.022]	−0.040 [0.032]	0.003 [0.027]	0.059 [0.017]***	−0.014 [0.013]	−0.040 [0.018]***
Reason for choosing job dummies	No	No	No	No	Yes	Yes	Yes	Yes
Observations	5,070	5,070	4,577	4,577	5,070	5,070	5,070	5,070
R ²	0.49	0.38	0.61	0.68	0.19	0.17	0.16	0.18

Notes: Unit of observation is a female survey respondent in a given post-MBA year. Individuals who had chil-

Table 10: reasons for not working, choosing jobs, and leaving jobs

6 Short summary

- The paper studies University of Chicago Booth MBAs from 1990–2006.
- Men and women have very similar earnings at graduation.
- The gender earnings gap grows rapidly and reaches nearly 60 log points a decade or more after MBA completion.
- Men take more finance classes and have slightly higher MBA GPAs; these differences matter because returns to MBA performance and finance coursework are large.
- Women have more career interruptions, and the earnings penalty to interruptions is large.
- Women work fewer hours over time, and the earnings structure of high-powered corporate and finance jobs heavily rewards long hours.
- Motherhood explains most of the labor-supply divergence between men and women.
- Women without children are much closer to men in experience and hours.
- Mothers with high-earning spouses reduce labor supply the most after childbirth.
- Fixed-effects event studies show sharp and persistent female labor-supply declines after first birth, but no comparable decline for men.
- The evidence does not show that MBA mothers are negatively selected on early predicted earnings.
- The results are an accounting of proximate mechanisms rather than a complete causal theory of preferences, discrimination, or social constraints.

7 Conclusion

7.1 Core conclusion

Bertrand, Goldin, and Katz show that the gender earnings gap among elite MBAs is not present at career start but grows quickly over the first decade. The main proximate forces are differences in MBA training,

career interruptions, and weekly hours worked. The largest dynamic force is motherhood, which is associated with reduced employment, reduced hours, and slower earnings growth.

7.2 Economic intuition

The relevant labor market strongly rewards continuous, long-hour, high-intensity work. Small deviations from that pattern have large earnings consequences. Because MBA mothers are more likely to interrupt careers and choose lower-hour or more family-compatible jobs, the earnings gap expands rapidly even among workers who begin with similar elite credentials.

7.3 Takeaway

The paper's lesson is that gender gaps among high-skilled professionals are dynamic. Equal initial access to elite education and entry-level jobs is not enough to prevent large later earnings differences. Career continuity, hours flexibility, job design, and household constraints become central as careers and families evolve.